

BCS1540 Algorithmic Design (4 ECTS)

Bachelor CS

Year 1

Period 5

Contact Information

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C4.008

Course Description and Outline

This course follows on from Data Structures and Algorithms, developing students' ability to design and (equally importantly) analyse algorithms for both correctness and efficiency. The course begins with a more in-depth treatment of greedy algorithms and dynamic programming (introduced in DSA), and how to analyse recursive algorithms with the master theorem. We then investigate how we can justify that a given problem is unlikely to have an efficient solution, via the notion of NP-completeness. Finally we cover some techniques we can use to attack NP-hard problems, namely backtracking, linear programming and branch-and-bound.

As well as lectures introducing the material, there will also be tutorials in which you can get help with concepts you are struggling with, try exercises (some of which will have model solutions presented during the tutorial), etc. – you are **strongly** encouraged to attend. At the end of the second tutorial there will be a 30-minute closed-book graded assignment in the same style as the final exam, covering the material from lectures 1-3 (i.e. greedy algorithms and the master theorem). This can count for 10% of the grade (see 'Grading' section below).

Course Materials

- Goodrich and Tamassia (2015) Algorithm Design and Applications. Wiley. ISBN: 978—1-118-33591-8

For more in-depth coverage:

- Cormen, Leiserson, Rivest, Stein (1990-2022), Introduction to Algorithms. MIT Press. ISBN 0-262-04630-X

Study Load

This course is worth 4 ECTS

- 1 ECTS = 28 hrs study
- 4 * 28 = 112 hrs
- 24 hrs lectures
- 10 hrs tutorial
- 112hrs-34hrs= 78 hrs general studying, review of lectures, revision etc (10 hrs/week). For example, this includes the time spent working on the problem sets related to each topic.

Final Exam

- The duration of the exam is 120 minutes, without breaks.
- The exam is closed book, and calculators will be neither required nor permitted.
- The exam will be a mixture of theoretical questions and algorithm design tasks, for which students will be required to write pseudocode but not Java code.

Resit

- There will be a resit final exam. The format will be identical to that of the final exam.

Grading

- The course grade will be determined by either the grade on the final exam, or a combination of 90% the final exam and 10% the assignment, whichever gives a higher grade. That is, the grade will be determined by the function $\max(\text{final exam grade}, 0.9 * (\text{final exam grade}) + 0.1 * (\text{assignment grade}))$.

Course Schedule

Date or week	Topic
31-03-26 Lecture 1	Introduction, greedy algorithms (SC)
02-04-26 Lecture 2	More greedy algorithms (SC)
07-04-26 Lecture 3	Master Theorem (DM)
09-04-26 Lecture 4	Divide and conquer (SC)
09-04-26 Tutorial 1	Greedy algorithms and the Master Theorem
14-04-26 Lecture 5	Dynamic programming: part 1 (SC)
16-04-26 Lecture 6	Dynamic Programming: part 2 (SC)
16-04-26 Tutorial 2	Divide and Conquer and Dynamic Programming, and graded assignment
21-04-26 Lecture 7	Problems we think we can't solve: the class NP (DM)
23-04-26 Lecture 8	NP hardness and reductions (DM)
28-04-26 Lecture 9	Backtracking and branch-and-bound (DM)
30-04-26 Lecture 10	Linear programming and ILP (DM)
30-04-26 Tutorial 3	NP hardness
07-05-26 Lecture 11	Slack
07-05-26 Tutorial 4	Backtracking/branch-and-bound/(I)LP
12-05-26 Lecture 12	Recap
12-05-26 Tutorial 5	Revision/troubleshooting

Information for Repeaters

- Network flows, approximation algorithms and randomised algorithms have been removed.
- Graded homework assignments have been replaced by in-tutorial assignment.